

A Lightweight CNN for Efficient Deepfake Detection of Low-resolution Images in Frequency Domain

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Abstract— Deepfake detection in frequency domain is extensively explored for two reasons – artifacts generated by the up-sampling layer of Generative Adversarial Networks are more prominent in the frequency domain and some image processing tasks are simpler and efficient in this domain. Traditional convolutional neural networks outperform the machine learning models for their ability to detect deepfakes of low-resolution images also. But these networks involve numerous trainable parameters resulting in huge computation cost, time and high memory requirements. This paper presents a lightweight Convolutional Neural Network (CNN) architecture that is optimized for efficient deepfake detection of low-resolution images in the frequency domain. The proposed model achieves almost the same accuracy as that of the traditional network but with the parameter requirement reduced by 92%. This renders the model suitable for use in memory-constrained and power-constrained environments such as mobile devices and other embedded systems.

Keywords—Deepfake detection, frequency domain, pixel domain, lightweight CNN, depthwise separable convolution

I. INTRODUCTION

The emergence of Generative Adversarial Networks (GAN) that can generate realistic synthetic data is considered a milestone in the advancement of machine learning and deep learning models, for its ability to surpass the creative thinking of humans. Interesting applications of GAN such as generating new patterns for the fashion industry, composing music and generating images from text account for its widespread popularity. Deepfakes are digitally manipulated visual or audio content to replace the likeness of one person with that of another and Deepfake technology leverages high-end systems and complex machine learning algorithms to co-create such convincing fake artistic contents which are indistinguishable from the real ones. However, the rapid growth of this technology has raised concerns regarding the potential for misuse and exploitation. The technology is being used to impersonate celebrities and spread fake news, commit financial fraudulence, perform election manipulation, identity theft, scams and hoaxes. The tremendous expansion

of social media has amplified the adverse effects of deepfake-based cybercrimes.

The potential misuse of deepfakes pose a growing threat to democracy, privacy and social security. Researchers, domain experts and policy makers work together to develop effective solutions to combat deepfake threats. Deepfake detection is one such solution that aims to develop techniques and tools to identify and distinguish deepfakes from real videos or images. This is important because deepfakes can be used to spread false information, demean individuals, and mislead public opinion.

Detecting deepfakes requires a combination of technical expertise, knowledge of machine learning algorithms, and access to large datasets of both real and fake videos or images. Owing to the fact that a few image processing operations are simpler and more efficient when applied in the frequency domain, some studies focus on doing deepfake detection in the frequency domain. Fig. 1 depicts the difference between authentic images and the fake images generated using deep neural networks and Fig.2 shows the representation of the real and fake images in different spectrums. The power spectrum is a 2D array of real-valued numbers that represents the strength of the frequency components. The frequency domain representation of a signal carries information about the signal's amplitude and phase at each frequency.



Fig. 1 Differences between Real and Fake Images (a) Real (b) Fake

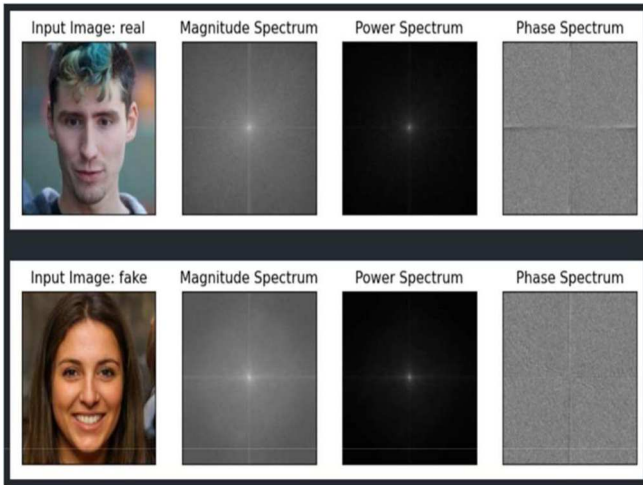


Fig. 2 Real and Fake Images in Different Spectrums

Existing literature demonstrates that in the frequency domain, traditional machine learning models are sufficient to successfully distinguish between authentic and fake images, for high resolution images; however, they are not efficient for low resolution images. Hence, it is necessary to build a robust image classifier using powerful neural networks such as CNN to accurately detect the artifacts left by the deep generative models. Recent studies focus on training convolutional neural networks in frequency domain to produce better results for low-resolution images too. But CNN incur excessive parameters resulting in huge computation costs, time and high memory requirements. This paper proposes and implements a lightweight CNN to make possible, the deployment of CNN on small devices. It requires fewer parameters, still without comprising the accuracy.

II. RELATED WORKS

In recent years, deep learning has made remarkable progress in every domain ranging from environmental modelling [1][2][3] to computer vision and robotics. Furthermore, digital face photos and video modification are of particular interest since they make use of the capabilities of GANs. Studies show that deep fakes are the outcome of Generative Adversarial Networks (GANs), which are two artificial neural networks that collaborate to produce a compelling realistic like media e.g., photos and videos [4],[5]. The Generative Adversarial Network is capable of producing a new portrait that approximates all thousands of photos it has been trained on earlier, without being an exact copy of any of them. This is achieved by utilizing two networks namely the generator and discriminator networks that are calibrated on the same dataset of pictures, video recordings, or audios. The first network attempts to generate new samples that are good enough to fool the second network, which works to determine whether the new media it sees is real. Different algorithms are used for the generation and detection of deepfakes [6]-[8]. Deepfakes creation methods have some challenges such as identity translation, spatial inconsistencies, temporal inconsistencies and more. According to [9], some artifacts are left by the network whilst creation which may not be visible to the naked eye but can be detected by image forensics and machine learning. Recent studies exploit these inconsistencies to detect the deepfakes successfully. Furthermore, Tariq, S et al. [10] and Ganiyusufoglu, I. et al

[11] were able to detect a deep fake media from a genuine one from behavioral aspects such as unusual head poses, mouth movement, unnatural blinking rate etc. As suggested in [12], video classification approaches can be used to take advantage of 3D input for the deepfake classification challenge. Except for Region-based Convolutional Neural Network (RCN), all of the convolutional networks they tested were pre-trained using the Kinetics dataset, a large-scale video categorization dataset. The methods were tested using video snippets of varying lengths. The authors chose the Celeb-DF (v2) dataset for their work, which contains 590 real films gathered from YouTube and 5639 comparable high-quality synthesized movies. Celeb-DF was chosen since it has shown to be a complex and challenging dataset for existing deepfake detection systems due to the lack of discernible visual artifacts in synthesized videos. Lewis, J. K et al. [13] go even a step further by including spectral information along with spatial and temporal data and propose a novel multileveled complex CNN-LSTM (Long Short-Term Memory) pipeline to detect Deepfakes. By utilizing both visual and audio artifacts, they were able to achieve high detection accuracy. As demonstrated in [14], a basic classifier following the frequency domain analysis of the images provides good accuracy even with a few labeled data and in unsupervised settings. Another study based on using the frequency spectrum to detect deepfakes has also shown 99% accuracy [15]. Another study by Frank, J., [16] experimentally proves that DCT transformed images are fully linearly separable, in contrast to classification on pixel domains that require neural networks. Kohli, A., & Gupta, A [17] have shown good results by converting the faces extracted from the videos to frequency domain and then feeding them to a 3-layered CNN to detect deepfakes.

In another unique approach, face manipulation is divided into face replacement and face reenactment, and face manipulation detection is treated as a three-classification problem [18]. They proposed the three-classification face manipulation detection technique (TFMD) based on attentional feature decomposition, as well as a novel face forgery feature representation. As suggested in [19], a biometric-based forensic technique can be efficiently used for detecting deep face-swap impersonations. Jung, T et al. [20] suggested an approach called Deep Vision that analyses eye blinks that are continually repeated in a very short period of time; Deep Vision is used as a measure to validate an anomaly based on the period, repeated number, and expired eye blink time. In seven out of eight different types of videos, their proposed methodology was able to correctly identify deepfakes (87.5% accuracy rate).

Deepfake techniques can only create face images with defined dimensions and low quality and these images must be further altered and blurred to match the original video's faces. This additive blur and ROI transformations leave unique artifacts in the generated deepfake movies, that may be efficiently recorded by spotting discrepancies between the Region of Interest (ROI) and the remainder of the image using Haar Wavelet transformation, that helps for deepfake detection [21].

III. METHODOLOGY

A. Dataset Description

Two different datasets are used : Flickr-Faces-High Quality dataset (FFHQ) consisting of 70000 images and Deepfake Detection dataset by Google consisting of 10000 real and fake images which are divided in the ratio 80:20 for training and testing, for achieving optimal results.

B. System Architecture

The proposed framework consists of two modules - Preprocessing and Deepfake detection.

a) Preprocessing

A robust pre-processing pipeline is employed to properly convert the videos into frequency data needed by the CNN. The pipeline is further divided into two phases – Face extraction and Transformation to frequency domain. Fig. 3 gives the schematic of the two phases of the preprocessing pipeline.

In the face extraction phase, the video is split into frames, and around 1/3 of frames per second are used to avoid redundant data. The frames are extracted using the *ffmpeg* tool, and then a face-recognition module, *dllib* is used to detect faces.

In the second phase, the detected faces are converted into the frequency domain using the 2D Global Discrete Cosine Transform (2D-GDCT). The 2D DCT is applied to a 256x256 image by first dividing it into 8x8 blocks, and each block is transformed using the 2D DCT to produce 64 DCT coefficients. The DCT coefficients that represent the frequency content of the image are stored in a compact and efficient way, in the form of a matrix, with the low frequency components in the upper left corner and the high frequency components in the lower right corner. Normalization is then performed to rescale the data to the range [0,1].

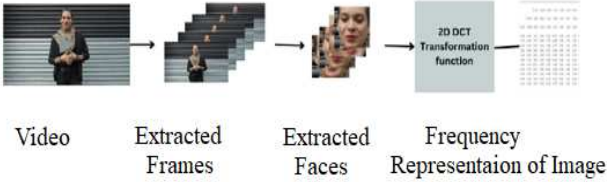


Fig. 3 Pre-processing (Feature Extraction and Domain Transformation)

Global Discrete Cosine Transform (GDCT) calculation is as follows.

Let F be the facial image of size $M \times N$, such that

$$F = [f(m, n), 0 \leq m \leq M-1, 0 \leq n \leq N-1]$$

and the GDCT of F be,

$$F_{gdct}(x, y) = a(x)a(y) \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) \frac{\cos(\pi x(2m+1))}{2M} \frac{\cos(\pi y(2n+1))}{2N}$$

where,

$$a(x) = \begin{cases} \frac{1}{\sqrt{M}} & x = 0 \\ \sqrt{\frac{2}{M}} & 1 \leq x \leq M-1 \end{cases},$$

$$a(y) = \begin{cases} \frac{1}{\sqrt{N}} & y = 0 \\ \sqrt{\frac{2}{N}} & 1 \leq y \leq N-1 \end{cases} \quad \dots (1)$$

b) Deepfake Detection

Traditional machine learning models fail to accurately distinguish between real and fake images of low resolution. CNN is a robust solution that works for both low and high resolution. The proposed architecture is a lightweight CNN optimized to work in small devices. It is implemented by replacing the normal convolutional layers with depthwise convolutions, and reducing the number of fully connected units carefully without much loss in accuracy score.

Depthwise Separable Convolution

Depthwise separable convolution is a type of convolutional neural network layer that consists of two consecutive layers - depth wise convolution and pointwise convolution.

In a traditional convolutional layer, the convolution is performed over the entire input volume, which includes all the input channels, using a filter that has a depth equal to the number of input channels. This can be computationally expensive, especially when dealing with large input volumes.

In a depthwise convolution, the convolution operation is performed separately for each input channel, using a filter that has a depth of one. This means that each filter is responsible for convolving only one channel of the input volume. This reduces the number of parameters required and makes the computation more efficient. After the depthwise convolution layer, a pointwise convolution layer is added to combine the output of the depthwise convolution layer. In a pointwise convolution, a 1x1 convolution is applied to the output of the depthwise convolution, to combine the filtered channels and produce the final output. This pointwise convolution helps to preserve the non-linearity of the model.

Depthwise separable convolution is obtained by combining the depthwise convolution and pointwise convolution layers. It is a more efficient alternative to traditional convolutional layers and has been used in many lightweight CNN architectures.

The proposed method uses a convolutional neural network to recognize deepfake faces in the target video. It is made up of six layers as illustrated in Fig. 4 - a depthwise separable convolutional layer, activation layer, additional pooling layer, Dropout Layer, Batch Normalization Layer and a fully connected layer. The feature extraction is handled by the convolutional, activation and pooling layers, while the fully connected layer transfers extracted features into classification output. As shown in Fig. 5, majority voting is applied to the results of the convolution network to classify the image into real or fake.

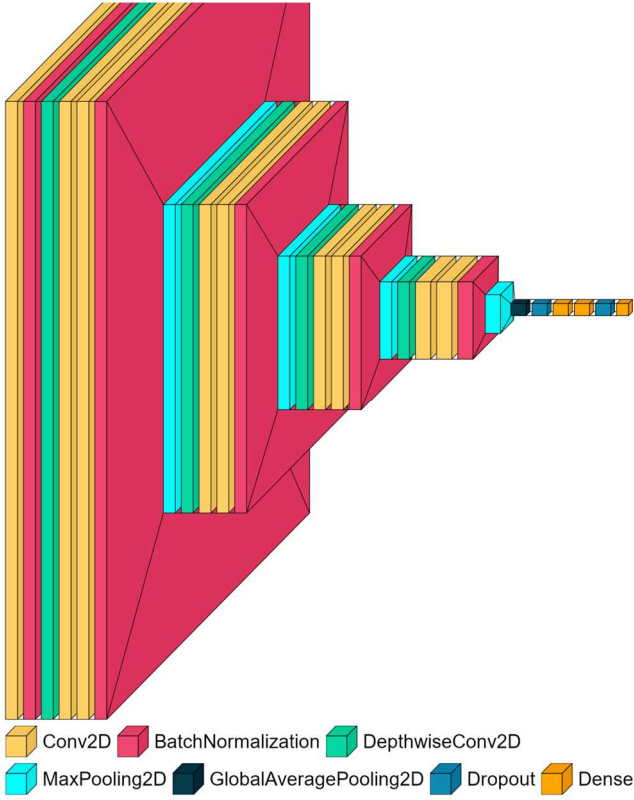


Fig. 4 CNN with Depthwise Separable Convolution Layers

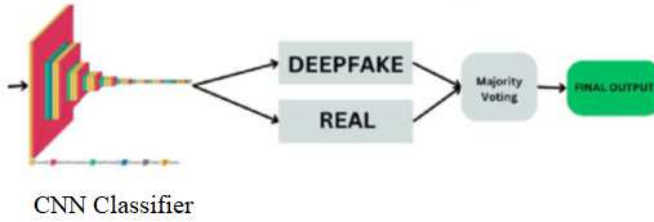


Fig. 5: Detection stage [Framewise Detection and Majority Voting]

C. Experimental Setup

The proposed model is implemented in Python and executed in a Dell Precision T7820 workstation configured with Intel Xeon processor, 96 GB RAM, and Nvidia Quadro RTX5000 graphics card.

From the Keras library, Conv2D layers with various filters such as 16, 32, 64, 128, and 256 are employed, with each followed by a BatchNormalization layer, a MaxPooling layer and a DepthwiseConv2D layer. At last, a GlobalAverage Pooling2D layer and a Dense layer are added. A learning rate of 0.0005 and Adam optimizer are used.

IV. RESULTS AND DISCUSSION

Traditional machine learning algorithms such as Logistic Regression, Support Vector Machine, Random Forest, Naïve Bayes are implemented for FFHQ Dataset High Resolution to establish the baseline performance. Table 1 presents the experimental results of these models in pixel domain and

frequency domain. Fig. 6 and Fig. 7 give the pictorial representation of these results.

From the empirical results for high resolution images that agree with [19], it can be inferred that traditional machine learning models are not much viable in pixel domain; nevertheless, they are able to differentiate high resolution deepfakes in frequency domain as their spectrum is clearly separable.

Table 1: Accuracy of Machine Learning Models in Pixel and Frequency Domain

MODEL	ACCURACY (Pixel Domain)	ACCURACY (Frequency Domain)
Random Forest	0.58	0.98
SVM	0.7	0.88
Naïve Bayes	0.68	0.96
Decision Tree	0.60	0.97
Logistic Regression	0.75	1

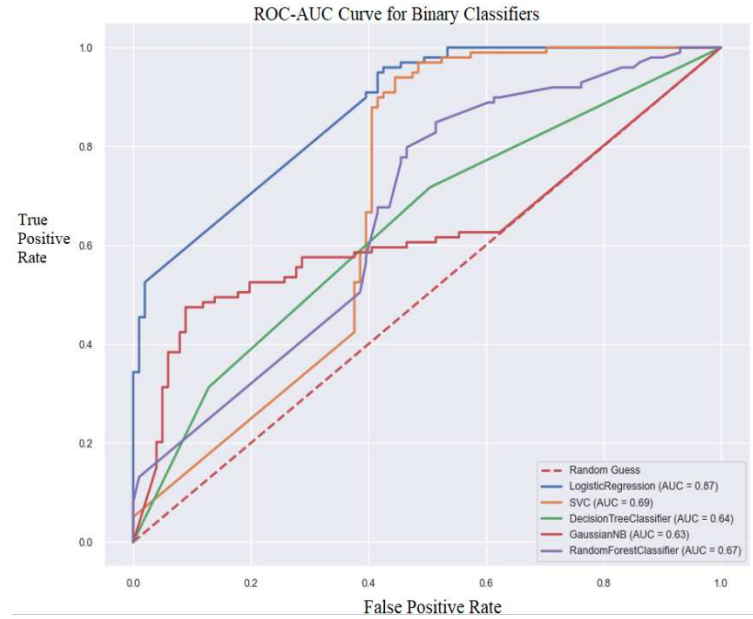


Fig. 6 Receiver Operating Characteristic-Area Under the Curve (ROC – AUC) Curve for Various Machine Learning Models in Pixel Domain

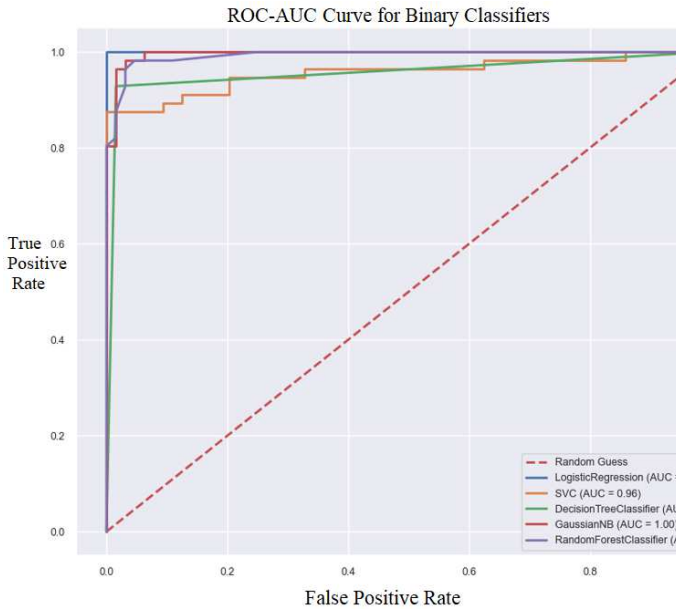


Fig. 7 ROC - AUC Curve for Various Machine Learning Models in Frequency Domain

A traditional CNN is implemented so as to detect the artifacts produced by GANs for low resolution images. The model is trained for 15 epochs with 20,000 training data and 2,000 validation data and the results of evaluation on both pixel and frequency domain are listed in Table 2. It can be observed that the model takes additional training time in frequency domain, owing to the domain transformation overhead but it has contributed to 4-5% increase in accuracy with only 20,000 training data. Fig. 8 and Fig. 9 show the comparative results of the traditional CNN model's accuracy in pixel and frequency domain.

Table 2: Accuracy and Training Time for Traditional CNN in Pixel and Frequency Domain.

METRIC	PIXEL DOMAIN	FREQ DOMAIN
Test Data Accuracy	85.85%	89.70%
ROC Score	0.8350	0.8860
Training Time (mins)	42	65

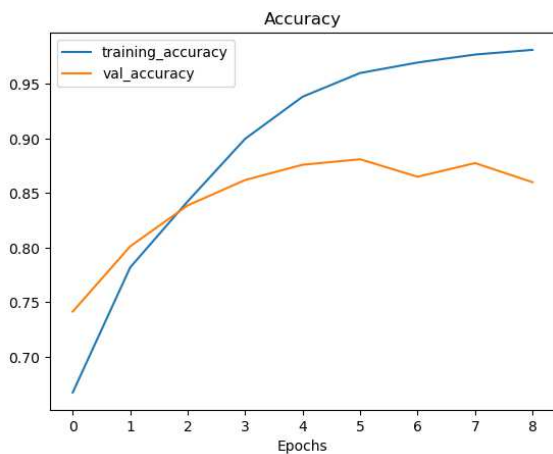


Fig. 8 Accuracy Curve for FFHQ Low-Resolution Dataset

in Pixel Domain

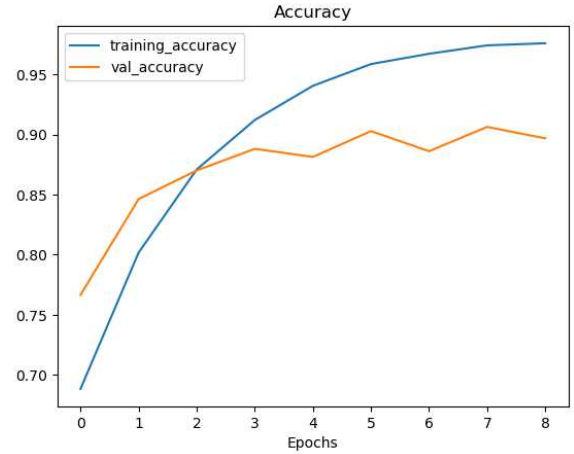


Fig. 9 Accuracy Curve for FFHQ Low-Resolution Dataset in Frequency Domain

The proposed lightweight CNN results in 84.27% accuracy which is only slightly less than the traditional CNN but with a huge reduction in the number of model parameters.

Let PT denote the number of trainable parameters for traditional CNN

Let PL denote the number of trainable parameters for lightweight CNN

Parameter reduction percentage is given by

$$\text{Parameter Reduction\%} = \frac{(PT-PL)}{PT} \times 100 \quad \dots\dots\dots (2)$$

The parameter reduction percentage for the proposed lightweight model with PL = 963,521 and PT = 12,502,801, is found to be 92.29%.

V. CONCLUSION AND FUTURE WORK

The fast-growing popularity of GANs and deepfake technology with their attractive applications has raised concerns regarding its potential misuse. It is indispensable to develop tools and techniques to detect deepfakes and combat the related cybercrimes. A lightweight CNN is designed and developed to perform deepfake detection for low-resolution images in frequency domain. For comparison purposes, various machine learning models such as SVM, Naïve Bayes and Random Forests and traditional CNN are also implemented. Results show that the machine learning algorithms are not suitable for working in pixel domain. Also in frequency domain, the accuracy is not great for low-resolution images. On the other hand, CNN works well for low-resolution images also but it incurs huge computation cost and time for training the excessive number of model parameters. The proposed lightweight CNN offers approximately the same accuracy as that of the traditional CNN but with 92% reduction in the total number of parameters required.

For future work, there are two primary objectives. First objective is to optimize the CNN model further to achieve higher accuracies for images and videos while reducing the number of trainable parameters. The second objective is to improve the model's generalizability to all existing methods of deepfake generation. This can be achieved by using a diverse dataset that encompasses a wide range of deepfake variations, including both known and emerging techniques.

By exposing the model to this comprehensive dataset, it can learn robust features and patterns that are indicative of deepfake content, enabling it to effectively detect deepfakes regardless of the method used to generate it. Iterative model evaluation and fine-tuning with diverse validation datasets should be performed. Additionally, exploring novel architectural designs, regularization techniques, and dataset augmentation approaches can further enhance the model's performance and generalizability.

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